

Elsai Mati Asefa

Lecturer and Researcher, Haramaya University

Personal Information

Date of Birth: January 16, 1999
Nationality: Ethiopian
Gender: Male
Email: elsyyymati@gmail.com; elsai.mati@haramaya.edu.et
Google Scholar <https://scholar.google.com/citations?user=Y1YDrd4AAAAJ&hl=en>
ResearchGate <https://www.researchgate.net/profile/Elsai-Mati-Asefa>
Telephone: +251921832876
Affiliation: School of Environmental Health, Haramaya University, Ethiopia
Present Address: 235, Harar, Harari, Ethiopia

Education

MSc in Ecotoxicology and Environmental Health

Hawassa University.
Oct, 2021 – Jul, 2023

Thesis Title: Towards Evidence-Based Pesticide Risk Assessment in Ethiopia: A Critical Evaluation of Existing Field Data and Regulatory Approaches.

Advisor: Dr. Mekuria Teshome Mergia

BSc in Environmental Health Sciences

Haramaya University.
Oct, 2016 – Mar, 2021

Thesis Title: Application of Geographic Information System in Solid Waste Landfill Site Selection in Harar City, Ethiopia.

Advisor: Yohannes Tefera Damtew

Other Qualifications and Training

- **Introduction and Access to Global Air Quality Forecasting Data and Tools:** Basics of air quality forecasting and how to access and interpret global air quality forecasts. NASA, Sept 30, 2021.
- **Occupational Safety and Health – EPHOC:** Current issues in occupational health and safety and preventive and control measures. Tulane University and the University of Alabama, April, 2021.
- **Basic Level Public Health Emergency Management Training:** Ethiopian Public Health Institute. 1/02/2021 – 5/02/2021.
- **Pedagogical skill training:** Teacher spoken discourse, classroom management and assessment. Haramaya University, 24/6/21 – 3/7/2021.
- **Emergency Preparedness and Response:** Analyze emergency risk to guide preparedness planning “Early Warning”. UNICEF, April, 2021.
- **Introduction to Monitoring and Evaluation:** Assessing the effectiveness and impact of programs, projects, and interventions. April, 2021.
- **Introduction to Water, Sanitation, and Hygiene (WASH) in Emergencies:** Key principles of WASH Promotion in emergency contexts. April, 2021.
- **Introduction to Infection Prevention and Control (IPC):** World Health Organization. April, 2021.

Teaching Experience

Lecturer

School of Environmental Health,
Haramaya University
Since August, 2023

Courses Taught (Undergraduate):

- Environmental Toxicology (EnvH4094)
- Air Pollution and Management (EnvH2081)
- Remote Sensing and Geographic Information System (GISc2096)
- Introduction to Environmental Health (EnvH2081)

Graduate Assistant

School of Environmental Health,
Haramaya University
May, 2021 – July, 2023

Instructing and advising undergraduate students

- Remote Sensing and Geographic Information System (GISc2096)

Laboratory Experience

Hawassa University Ecotoxicology Lab (2022-2023)

- Conducted physicochemical and bacteriological quality analyses of surface water and wastewater.
- Collected tap water from Hawassa city and analyzed it for heavy metal contents using atomic absorption spectrometry.
- Collected water and fish samples from Lake Hawassa and analyzed them for microplastic occurrence and abundance using appropriate techniques.
- Collected fish samples and analyzed them for *DDT* concentrations using GC-ECD, and extrapolated the findings for human exposure and health risks.

Haramaya University Environmental Health Skill Lab (2018-2021)

- Collected water samples from Lake Haramaya, Lake Adelle, and household taps, and analyzed them for bacteriological water quality.
- Collected wastewater samples from Harar city and analyzed them for BOD content.
- Conducted passive air sampling for the analysis of indoor bacterial and fungal loads.
- Experimented with the preparation of agar media using techniques such as membrane filtration and multiple tube fermentation.

Research Experience

I have always been passionate about science, and my journey into research and publication began during my **bachelor's studies, deepening further during my master's**. I have participated in various research projects, laboratory investigations, and environmental modeling, and have presented my findings on multiple occasions.

In my bachelor's studies, I focused on **environmental modeling**, utilizing **GIS** to identify a suitable landfill site in Harar city. I also contributed a book chapter titled "Application of GIS in Solid Waste Management." Additionally, I engaged in **systematic reviews and meta-analyses**. My research interest evolved when I joined Hawassa University for my master's, where I dedicated myself to studying *chemical exposure and its effects on human and environmental health*. My thesis explored **pesticide use, risks, and regulatory approaches** in Ethiopia. I implemented a **novel** method combining evidence mapping, systematic review, meta-analysis, and modeling. I compiled existing field concentrations from literature to create a database on pesticide concentrations in Ethiopian surface waters. Moreover, I collected data for pesticide fate and transport modeling, successfully simulating pesticide concentrations using the **PRIMET model**.

I have published **10 research papers and 1 book chapter**, with some appearing in Scopus **TOP5** and **TOP1 journals**, totaling 86 citations (**h-index 4**; Scopus [3]). As an academic staff member at the School of Environmental Health, Haramaya University, I actively engage in research, write and apply for internal grants, and design community service projects. I have also **advised** several final-year **undergraduate research projects** and provided expert guidance to other research staff.

Overall, these are the skills and experiences I can bring to your lab and research team:

- I am well-experienced in using secondary chemical exposure data and analyzing it to better understand exposure status (multiple, mixture) and health risks (acute, chronic, disease-specific) among the general population and vulnerable groups, following US EPA and EFSA.
- I am also experienced in simulating the fate and transport of pollutants in the environment. I am familiar with **FOCUS models** and other multimedia models such as the **USEtox** model, as well as watershed models like **SWAT**.
- I have used various chemical properties and effect databases, such as **PPDB, ECOTOX, and IRIS**, to derive substance-specific data. I have derived threshold levels using *dose-response* data for various endpoints and species.
- I have experience with geospatial analysis using **QGIS/ArcGIS** software and advanced statistical models using **R** and other statistical software.

Research Summary

A Bibliometric, Systematic Review, and Meta-analysis

- Systematically designed search strategy, evidence synthesis, and analysis following PRISMA and Cochrane guidelines in both Ethiopian and global contexts. Some of my works are published in well-recognized journals and are highly cited. During my MSc studies, I employed a novel method that combined evidence mapping, systematic review, meta-analysis, and modeling to create a comprehensive database of pesticide concentrations in Ethiopian surface waters.

Environmental Modeling and GIS Applications

- Applied GIS to various areas of Environmental Health, particularly solid waste management. Published an original article and a book chapter on the application of GIS and multicriteria decision-making techniques in the suitability analysis of landfill sites. Used spatially explicit environmental models to predict the fate and transport of pesticides in the environment, which were used for human and environmental risk assessments.

Chemical Exposure and Environmental Health

- My MSc research focused on investigating chemical exposures and their adverse effects on human and environmental health, particularly pesticides. I analyzed pesticide use, associated risks, and regulatory approaches in Ethiopia. I collected existing field concentrations from the literature and created a database on pesticide concentrations in Ethiopian surface waters. I also successfully simulated pesticide concentrations in Ethiopian waters using the PRIMET model.

Pesticide and Heavy Metal Risk Assessment

- I analyzed the acute aquatic ecosystem risks of pesticides in Ethiopia using field-collected data and relevant toxicological threshold levels, published in the **Science of The Total Environment**. Focusing on persistent organochlorine pesticides, I estimated potential bioaccumulation and bioconcentration factors and human health risks from DDT intake through water ingestion and fish consumption, published in **Toxicology Reports**. Combining actual field and simulated pesticide concentrations in Ethiopian waters, I calculated the pollution status using a novel scoring approach across 22 site-specific locations and estimated acute and chronic health risks for 99 individual pesticides out of the 300 pesticides screened, published in the **Journal of Hazardous Materials**. Adopting a similar approach, I also conducted the same analysis for heavy metals detected in Ethiopian drinking water sources, published in the **Science of The Total Environment**.

Research Interest

My research interests lie at the intersection of **environmental science, chemical exposure, and public health**. Addressing the intricate relationship between environmental pollution and adverse effects, my goal is to incorporate **chemical exposome** into my research. In this regard, I am highly interested in *High-throughput Non-targeted Exposomics, Omics Technology, Fate Modeling, Geospatial Epidemiology, Wastewater Fingerprinting, and Risk Assessments*. Utilizing a combination of these methodologies, my aim is to identify and characterize chemicals in different environments, food products, and consumer goods, and link them with health impacts, including population-based exposure influencing factors, at varying **spatiotemporal** scales. I am also interested in incorporating **exposome data into modeling** to facilitate the simulation and prediction of health outcomes. Particularly, my focus is on underrepresented, highly exposed, and vulnerable low- and middle-income countries, mainly **sub-Saharan Africa**. Moreover, I am keen on expanding my research network through multidisciplinary collaborations with global experts and institutions, contributing to One Health, Planetary Health, and Sustainable Development Goals.

Research Publications

*Corresponding Author

1	Elsai Mati Asefa* , Yohannes Tefera Damtew, Józef Ober (2024). Pesticide Water Pollution, Human Health Risks, and Regulatory Evaluation: A Nationwide Analysis in Ethiopia. <i>Journal of Hazardous Materials</i> .	https://doi.org/10.1016/j.jhazmat.2024.135326
2	Elsai Mati Asefa* , Yohannes Tefera Damtew, Dechasa Adare Mengistu, Sina Temesgen Tolera, Faye Fekede Dugusa, Abdi Berhanu, Berhan M. Teklu, Ermias Deribe Woldemariam (2024). Heavy metals in Ethiopian drinking water and public health risks: Insights from nationwide and regional analysis. <i>Science of the Total Environment</i> .	https://doi.org/10.1016/j.scitotenv.2024.174527
3	Elsai Mati Asefa* , Mekuria Teshome Mergia, Yohannes Tefera Damtew, Dechasa Adare Mengistu, Faye Fekede Dugusa, Roba Argaw Tessema, Jerry Enoe, Józef Ober, Berhan M. Teklu, Ermias Deribe Woldemariam (2024). Organochlorine Pesticides in Ethiopian Waters: Implications for Environmental and Human Health. <i>Toxicology Reports</i> .	https://doi.org/10.1016/j.toxrep.2024.06.0017
4	Elsai Mati Asefa* , Mekuria Theshome Mergia, Shiferaw Ayele, Yohannes Tefera Damtew, Berhan Mellese Teklu, Ermias Deribe Woldemariam (2024). Pesticides in Ethiopian surface waters: A meta-analytic based ecological risk assessment. <i>Science of the Total Environment</i>	https://doi.org/10.1016/j.scitotenv.2023.168727
5	Jerry Enoe, Michael Sutherland, Dexter Davis, Bheshem Ramlal, Charisse Griffith-Charles, Keston H. Bhola, Elsai Mati Asefa* (2024). A conceptional model integrating geographic information systems (GIS) and social media data for disease exposure assessment. <i>Geospatial Health</i>	https://doi.org/10.4081/gh.2024.1264
6	Hafso Abdurahman Ibrahim, Abdi Wariyo, Elsai Mati Asefa , Abera Cheru, Arega Abebe Lonsako, and Gebisa Dirirsa (2024). Measles second dose vaccine uptake and associated factors among under-five children in Jigjiga City, Somali Region, Eastern Ethiopia: a community-based cross-sectional study. <i>Frontiers in Public Health</i>	https://doi.org/10.3389/fpubh.2024.1395802
7	Elsai Mati Asefa* , Mekuria Teshome Mergia (2022). Human exposure to indoor air pollution in Ethiopian households. <i>Heliyon</i>	https://doi.org/10.1016/j.heliyon.2022.e11528
8	Dechasa Adare Mengistu, Gebisa Dirirsa, Elsai Mati Asefa , Dinku Mekbib Ayele, Kefelegn Bayu, Wegene Deriba, Fekade Ketema Alemu, Yohannes Mulugeta Demmu, Yohanis Alemeshet Asefa, and Abraham Geremew (2022). Global Occupational Exposure to Blood and Body Fluids among Healthcare Workers: Systematic Review and Meta-Analysis. <i>Canadian Journal of Infectious Diseases and Medical Microbiology</i>	https://doi.org/10.1155/2022/5732046
9	Elsai Mati Asefa* , Yohannes Tefera Damtew, Kefelegn Bayu Barasa (2021). Landfill Site Selection Using GIS Based Multicriteria Evaluation Technique in Harar City, Eastern Ethiopia. <i>Environmental Health Insights</i>	https://doi.org/10.1177/11786302211053174
10	Elsai Mati Asefa* , Kefelegn Bayu Barasa, Dechasa Adare Mengistu (2022). Application of Geographic Information System in Solid Waste Management. <i>Book Chapter: Geographic Information Systems and Applications in Coastal Studies</i>	https://doi.org/10.5772/intechopen.103773
11	Temesgen, Tolera Sina, Deressa Alemayehu, Berhanu Abdi, Abdeta Tilahun, Elsai Mati Asefa , Kitessa Monas, Balis Bikila, and Firdisa Dawit (2023). Toxicological Distribution of Mercury Concentrations in Abiotic and Biotic Environments: A Systematic Review and Meta-Analysis. <i>East African Journal of Health and Biomedical Sciences</i>	http://ejol.ethernet.edu.et/index.php/EAJHBS/article/download/2752/1495

Conferences and Presentations

1	Elsai Mati Asefa , Mekuria Mergia Teshome. Assessing the Impacts of Pesticide Use on Freshwater Ecosystems in Sub-Saharan Africa: Insights from Agroecosystems in Ethiopia (<i>Abstract Accepted</i>). <i>GEST-2024 conference</i> , https://environment-conference.com/experts	<i>Virtual Presentation.</i> London, UK 04-06 November, 2024
2	Elsai Mati Asefa . Regulatory risk assessment of pesticides and environmental protection in sub-Saharan Africa: insights from Ethiopia. <i>Cell Symposia: A circular economy for the chemical sector</i>	<i>Poster presentation.</i> Wales, UK. July 22-24, 2024.
3	Elsai Mati Asefa . Pesticide Use and Risks in Ethiopia: A Bibliometric Analysis, Systematic Review, and Meta-Analysis	<i>Seminar Presentation.</i> Hawassa University. Jan 28, 2023

Honours and Awards

- 1

Master’s Degree Sponsorship by Haramaya University: Covers tuition fee, cost of living and research funding of 25,000 ETB (twenty-five thousand Ethiopian birr). Haramaya University (2021 - 2023).
- 2

Certificate of Recognition for Successful Implementation of Community-Based Projects. Aboker Woreda, Harari Regional State (Feb, 2021).
- 3

The scholarship for historically excluded trainees to attend Cell Symposia: A circular economy for the chemical sector. 22-24 July 2024, Cardiff, Wales, UK.
- 4

Certificate Of Presentation. Cell Symposia: Circular Chemistry Certificate of Presentation (July, 2024).
- 5

Certificate of Reviewing: In recognition of the review contributed to Science of the Total Environment journal (June, 2024).

English Proficiency

IELTS: 24ET501004ASEE001A (July 01, 2024)

Overall: 6.5; Listening: 7, Reading: 6.5, Writing: 6.5, Speaking: 6

Other Skills

Statistical Software: R (*advanced user*), STATA, SPSS

GIS Software: QGIS, ArcGIS

Microsoft Office: Microsoft Word, Excel (*advanced user*), PowerPoint, Access

Evidence Synthesis: Cochrane, PRISMA, RevMan

References

Dr. Mekuria Teshome (Assistant Professor) mekuriamergia@gmail.com	Dr. Ermias Deribe (Associate Professor) erimdw@gmail.com	Dr. Roba Argaw (Assistant Professor) robaargaw@gmail.com
Yohannes Tefera (PhD candidate) yhnnstefera@gmail.com	Sina Temesgen (PhD candidate) sinatem3@gmail.com	